

Arborescent Eternalism: An Objective Modal Structure for the Block Universe

Architecture, Interactions and Variational Actualisation

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Abstract: This paper introduces Arborescent Eternalism, a framework that extends the block universe model by incorporating a structured space of dynamically admissible histories. Three standard approaches frame the relation between spacetime and possibility: the block universe, branching accounts and modal realism. Arborescent Eternalism (AE) proposes a fourth approach: an explanatory architecture combining one tenseless actual history with an objective organisation of physically admissible alternatives. A rooted event structure represents dependence, conflict, independence and provenance; an independently specified projection relates its physical realisation to spacetime. Reference weights describe typicality, while a distinct selection rule concerns actuality. Agents and interacting systems belong to complete histories rather than acting from outside the block. Compatible local developments can compose, but incompatible alternatives are not thereby interacting worlds. The research programme connects three explanatory tasks: grounding physical orientation in structural extension, retaining whole-history admissibility beyond finite descriptions, and explaining local recording readiness and typical macroscopic entropy growth. Companion studies establish conditional results for each task. In the recording model, reversible recording capacity constrains preparation, and the same prepared device exhibits typical macroscopic entropy growth relative to a physically specified collective readout, without requiring global minimisation. AE's proposed contribution is the reuse of a common modal architecture across these explanations. The results do not derive cosmological initial conditions or establish AE's unique ontology. The outstanding test is a common physical realisation that satisfies the constraints of all three studies without fitting a separate model to each explanatory target.

Keywords: eternalism; explanatory underdetermination; objective modality; temporal orientation; agency; projective weights; entropy; global constraints; actualisation

1 From an underdetermined block to the AE proposal

1.1 *The modal question left open by a complete block*

Eternalism represents the world as one four-dimensional history in which past, present and future events are equally real. The block representation is familiar in B-theories of time and fits naturally with relativistic spacetime, although relativity alone does not settle the metaphysics of time. That claim concerns the temporal status of events. It does not by itself specify the modal structure surrounding the realised history: which other complete histories are physically admissible, how they are related, how they are weighted, or what distinguishes actuality from admissibility.

This is not an objection to relativity. Relativistic physics constrains spacetime geometry and the relations among events, while concrete physical theories add dynamical laws, solution spaces, boundary conditions and measures. Those additions may determine a rich space of admissible models, but their modal interpretation is not supplied by the bare assertion that one block is realised. The question is therefore interpretive and structural: how should a tenseless theory represent physically constrained alternatives without turning them into further realised worlds?

Several positions should be kept apart. Alternatives may be treated as merely epistemic; represented by branching structures intended to model objective indeterminism (Belnap, 1992); interpreted as emergent worlds in Everettian quantum mechanics (Everett, 1957; Wallace, 2012); or treated as concrete possible worlds in Lewisian modal realism (Lewis, 1986). AE proposes an enrichment of a B-theoretic block: it retains one concretely realised complete history while treating the organisation of admissible alternatives as objective structure.

The realised history may already be complete. What it does not contain by itself is the counterfactual and nomological structure needed to compare that history with admissible alternatives. A theory can encode this information through laws and a solution space, but it cannot both use that structure and claim that the realised manifold alone performs all the explanatory work.

AE begins from this modal underdetermination. Its task is to state the additional structure explicitly, keep its components from doing one another's work, and test links between temporal orientation, whole-history admissibility, local physical mechanisms and actuality.

A complete block tells us what happens in the realised world. AE asks what further structure is required to represent what could physically have happened and why one admissible history is actual. The proposal is deliberately compatible with a tenseless ontology: neither modal extension nor selection is a process occurring outside or above spacetime.

The standard resources of physical theory already include laws, solution spaces, probabilities, boundary conditions and counterfactuals. AE does not deny those resources or claim novelty for each of them. It proposes a typed architecture that separates their roles and imposes explicit bridges between them. If an enriched B-theory reproduces those roles and bridges, it contains functionally equivalent structure even if it rejects AE's vocabulary.

This makes the programme assessable. AE earns an explanatory advantage only if components introduced for one reason also constrain other results without being retuned for each target. Otherwise it would merely collect several independent models under one name.

1.2 The foundational thesis

AE's foundational thesis is that a tenseless world can contain one actual history and an objective organisation of alternatives without multiplying concrete worlds. The organisation must record which events depend on others, which alternatives conflict, which independent developments may coexist or reconverge, and how physically admissible configurations are represented on the realised spacetime.

'Objective' means that admissibility is not reduced to an observer's ignorance. 'One actual history' means that AE does not require every admissible history to be concretely realised. The non-actual alternatives may be understood as objective physical possibilities or

as nomological structure. Minimal AE therefore lies between purely epistemic possibility and a concrete multiverse.

Nothing in this account happens in a second time. The root is not a moving present, and the actuality rule is not a later event choosing among already completed universes. All physical events, including deliberations and decisions, belong to the complete tenseless history. Rootedness, projection and selection express structural relations within the model.

The formalism is not claimed to be uniquely indispensable. What the programme requires is the structural content it encodes. Removing rooted extension loses the programme's source of orientation; removing objective alternatives eliminates the domain of comparison for weights and selection; removing provenance permits incompatible pasts or duplicate descriptions to be conflated; and removing physical projection breaks the link between modal organisation and spacetime. An equivalent formalism may redescribe these functions, but it cannot remove them while retaining all of the programme's explanatory results.

The minimal conceptual core is M , E , Π and a distinguished history H^* . Reference weights and J belong to the variational formulation; A is an additional agency hypothesis. The components have the following roles:

M is the common tenseless physical base (the common spacetime manifold) on which the realised spacetime history is represented.

E is the rooted modal structure, arborescent in its unfolding, organising dependence, conflict, independence and the provenance of alternatives;

Π maps physically admissible configurations of E to M . It is fixed independently of selection and supplies the bridge through which modal structure can acquire physical significance.

$\mathcal{H}$ is the resulting set of physically admissible complete histories.

w , or equivalently a reference measure v_0 where appropriate, assigns fixed typicality weights to admissible alternatives.

In the variational formulation, J is the modal variational functional on complete admissible histories, with E , Π and w specified beforehand.

S denotes the rule by which the variational formulation distinguishes the actual history H^* . The general physical form of J and the conditions for a unique resolution remain to be established.

An argmin is one provisional implementation, not a derived actuality law.

A represents agentic information where agents are modelled. It may contribute to J and alter which history is selected, but in the present theory it neither creates E or Π nor remodels w .

1.3 *How the argument is supported*

Sections 2 and 3 define the ontology and dependency order. Sections 4 and 5 state orientation, record and weighting results. Sections 6 and 7 explain agency, physical interactions and composition. Section 8 distinguishes the local entropy mechanism from global selection. Sections 9 to 11 present the research programme, its comparative burden and the outstanding physical bridges.

The companion papers are Temporal Directedness Without an Open Future (Mury, 2026b), Global Constraints Beyond Finite Modal Data (Mury, 2026c), and Recording Capacity and Macroscopic Entropy in Arborescent Eternalism (Mury, 2026d), hereafter the orientation, global-admissibility and recording studies. The foundational paper supplies their common ontology and explanatory roles. Their current results are conditional sector tests; a common physical realisation remains a further research obligation.

2 **Objective alternatives and one realised world**

A tenseless theory must distinguish three questions: which complete histories are physically admissible, how much reference weight different sectors carry, and which history is actual. This section introduces those distinctions in that order and then states the provisional variational implementation used in the constructive models. The point is not that every theory must use an argmin rule, but that no weighting measure by itself constitutes an actuality rule.

2.1 *What AE adds to eternalism*

Write $\mathcal{H}$ for the set of physically admissible complete histories and E for the rooted event structure from which those histories are built. E is not a collection of parallel realised universes. It records dependence, incompatibility, independence and provenance before one asks which complete history is actual.

Not every mathematically describable continuation is physically admissible. The relevant physical laws determine which configurations belong to E_{phys} and which complete histories belong to $\mathcal{H}$. AE does not manufacture those laws; it requires the admissibility distinction to be objective and explicitly represented (Le Bihan, Margoni, & Vieser, 2025).

The structural projection $\Pi: E_{\text{phys}} \rightarrow M$ then specifies how admissible modal configurations are represented on the common tenseless physical base. Π is defined before the realised history is selected. It is neither a probability measure nor an outcome of J . This separation allows the same event structure to organise possibilities while physical theory determines which of them receive a coherent spacetime representation.

Those laws need not always work as a step-by-step recipe from an earlier state to a later one. They may constrain a whole history at once. This is why all-at-once approaches to laws are relevant to AE: they make it coherent to speak of global admissibility and global selection without introducing a second temporal process (Adlam, 2022).

For example, a whole-history rule may require that an event occur at least once. Every finite description in which it has not yet occurred can still extend to an allowed history. The complete history in which it never occurs is nevertheless excluded. Finite possibilities therefore need not determine global admissibility. The global-admissibility study establishes the conditions under which finite restrictions do suffice (Mury, 2026c).

2.2 Possibility, typicality and actuality

Let $\mathcal{H}$ be the set of admissible complete histories. Let ν_0 be a reference measure on suitable subsets of $\mathcal{H}$, let $B \subseteq \mathcal{H}$ be measurable with $\nu_0(B) \in [0,1]$, and let $H^* \in \mathcal{H}$ be the one realised history.

$$H \in \mathcal{H}, \quad \nu_0(B), \quad H = H^*.$$

The distinction is crucial. Membership in $\mathcal{H}$ answers “is this history allowed?”; $\nu_0(B)$ answers “how much reference weight does this sector carry?”; H^* answers “what is actual?”. A rare sector can therefore contain the actual history without becoming less rare after the fact.

2.3 Global evaluation, variational selection and no becoming

In earlier constructive variational implementations, J is evaluated over complete admissible histories, and the provisional rule identifies H^* with a global minimiser under separate existence conditions. Π and the reference weighting are fixed beforehand. This is a candidate implementation, not the definition of the minimal conceptual core or a law derived from it.

A constructive variational implementation can separate reference surprisal, physical compatibility and goal-sensitive compatibility. Section 2.4 states a finite-model candidate. These terms do not determine the selection semantics by themselves, and none of them guarantees entropy growth along a selected history. Global evaluation is compatible with a tenseless conception of law (Adlam, 2022; Chen & Goldstein, 2022).

Why keep J and S separate? Because assigning weights to histories does not by itself make one of them actual. In a concrete model, H^* may be the unique minimiser of J ; in another model, J may only define a probability law and a further account of actuality is still needed. The variational idea is therefore a candidate mechanism for global selection, not a theorem that every physical world minimises the current J .

The variational formulation must further specify whether actuality is fixed by stationarity or by global minimisation. A stationary solution can be a minimum, a maximum or a saddle; replacing stationarity by global minimisation requires additional assumptions. The existence of one actual history is an ontological commitment, not a uniqueness theorem already proved for J .

2.4 A minimal constructive candidate for history selection

The architecture now needs an evaluator that keeps reference typicality, physical compatibility and agentic compatibility separate. A minimal complete-history candidate with that dependency order is

$$J(H \mid E, \Pi, w, A) = -\log w_E(H) + B_{E, \Pi}(H) + C_A(H)$$

with the provisional selection rule

$$H^* \in \text{Argmin}_{H \in \mathcal{H}} J(H \mid E, \Pi, w, A)$$

The terms have different jobs: reference surprisal, a whole-history physical compatibility cost and a goal-conditioned compatibility cost. In the finite witnesses, physical cost is defined independently of an entropy reward. Neither A nor its compatibility term changes w . This decomposition is a candidate for testing the variational formulation, not a claim that nature has exactly three fundamental addends.

Earlier finite generative witnesses instantiate such an evaluator and permit goal changes with the non-agentic data held fixed. They test constructive consistency and sensitivity, not the physical truth or uniqueness of J . Their detailed enumerations remain supplementary results; they are not the central claim of the current recording paper. The reference-calibrated witness is archived at https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/computational/j_reference_calibrated_v6_73/ (README.md, j_reference_calibrated_gate.py and results.json).

The point is that goal-sensitive selection can be represented without retrospectively changing the reference law. A realistic application still has to justify the physical feature map, the role of A and the selection rule independently. In a continuous history space, a measure need not assign positive weight to individual histories; the displayed surprisal candidate requires atoms or a separately specified density/reference construction.

3 The architecture of AE

This section assembles the components introduced so far and fixes their dependency order. It explains what E , Π , w , J , A and S each contribute, why projection precedes evaluation, and why reconvergent descriptions must not be counted as new physical possibilities. The aim is to prevent the later results from borrowing explanatory work across layers without an explicit bridge.

3.1 Events, configurations and projection

The following signature condenses the ontology and its interfaces. It should be read as a typed map of roles, not as a list of independent substances:

$$AE = (M; E; \Pi; H^*; w; J; A; S).$$

The formula above is a map of the theory, not a generative sequence and not a claim that every symbol is a new substance. M is the physical base, E organises alternatives, Π relates physically admissible configurations of E to M , H^* is the realised history, w or v_0 supply reference weights, J evaluates complete alternatives, A represents agentive constraints where present, and S denotes the rule that distinguishes actuality. The foundational order is $E \rightarrow \Pi M$, followed by selection within the already projected admissible history space; it is not $J \rightarrow \Pi$ or $A \rightarrow w$.

M is the common tenseless physical-event base. AE does not claim that the bare combinatorial form of E by itself generates spacetime, nor that a variational extremum derives Π . The projection Π is a structural datum of the model: it relates the physically admissible part of E to M and may, in a concrete implementation, be constrained by independently stated physical laws. This relation is not a temporal process by which spacetime comes into existence and is not an outcome of selection. Figure 1 should therefore be read as a representation of structural dependence, not as a claim that the shape of the event tree alone makes M emerge. For concurrent configurations, a physical anchor must also be specified: using the last event of an arbitrary serialisation would not define a projection on the configuration quotient. The general covariant construction is an open physical bridge, not supplied by the bare event graph.

To represent alternatives without forcing them into a strict tree, AE uses an event structure E . Write $\mathcal{C}(E)$ for its coherent finite configurations. Dependence says that one event requires another; conflict says that two events cannot belong to the same history; independence says that two events can coexist in either order. This is why compatible paths may reconverge. The root/anchor supplies the common point from which extension is organised (Nielsen, Plotkin, & Winskel, 1981).

$$\Pi: E_{phys} \rightarrow M.$$

Write E_{phys} for the physically admissible part of E .

The projection is $\Pi: E_{phys} \rightarrow M$ and is defined before the selection of H^* , not only on H^* . Some describable branches of E may fail the physical admissibility conditions and therefore have no image in M . A complete admissible history must project coherently across the physical base. In plain language: E describes the structured menu of possibilities; Π fixes how physically admissible possibilities are represented on M ;

J and S then concern which complete projected history is realised.

Π is neither a probability measure nor a quantity varied by J in the present theory.

3.2 *Arborescent organisation, projection and reconvergence*

The word “arborescent” therefore should not be read as “a mathematical tree with no mergers”. Incompatible alternatives remain distinct because merging them would erase which past occurred. Independent developments are different: they can be represented in different orders and later correspond to the same physical configuration. AE allows that benign reconvergence.

This matters quantitatively. If two drawings are merely two orders of the same independent events, counting them as two physical possibilities would create fake probability and fake entropy. AE therefore counts physical alternatives, not presentation duplicates.

Projection must consequently respect the configuration quotient and the physical anchors of the event realisation, rather than be reconstructed from whichever path was drawn first. This prevents a convenient picture of branching paths from being mistaken for the physical structure itself.

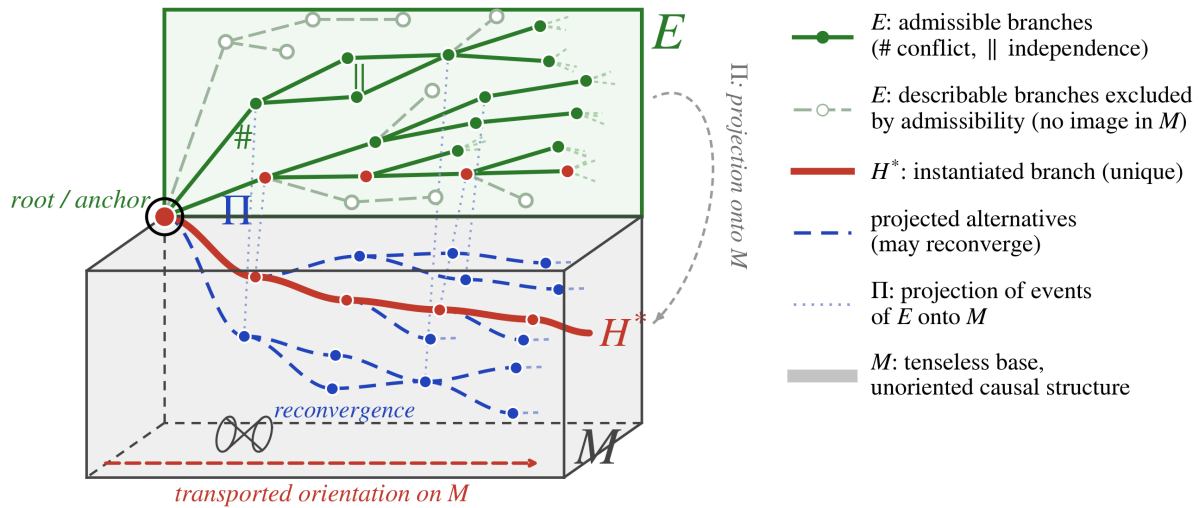


Figure 1. Schematic AE architecture. Green branches represent physically admissible alternatives; pale branches are excluded. Conflict (#) preserves incompatible provenance, while independent developments (||) can reconverge. Π relates the physical event realisation to the tenseless base M . The red curve denotes H^* , the one concretely realised history in the minimal interpretation. Blue curves denote projected alternatives. The arrow illustrates orientation transport under the conditions of Section 4; it is not an external time or an entropy gradient. Weights, agency, evaluation and selection are omitted.

3.3 A map of explanatory roles

The event-structure notation is not uniquely indispensable. The programme tests what is lost when a specified role is removed: extension organises orientation, conflict and independence organise alternatives, provenance preserves incompatible pasts, and projection provides physical meaning. An equivalent formalism can retain those roles. Establishing their reuse in one independently fixed physical model is a further obligation.

Component	Object or operation	Role in the proposed explanation
$\mathcal{C}$ and E	Admissibility and organised events	What is possible; dependence, conflict and independence.
J	Global evaluation / variational sector	Evaluates complete alternatives in a variational formulation; a selection rule remains separately specified. Does not imply entropy growth.
w and v_0	Coherent weights and reference measure	Fixed reference weight, typicality and structural entropy
A	Agentive variables and goal constraints	Goal-sensitive contribution to selection; does not set or remodel w , and is not a source of temporal orientation

Component	Object or operation	Role in the proposed explanation
Π	Structural event projection	Fixed relation from physically admissible modal configurations to the physical base; not generated by J
e_Σ, μ_Σ	Physical evaluation and measure	Phase states and physically measured macrovolumes
S and H^*	Selection structure and distinguished history	Actuality within the tenseless organisation

J evaluates complete alternatives in the variational formulation. It is not automatically a microscopic action or a law of actuality. Earlier restricted constructions give sufficient conditions for low-sector selection; they do not guarantee subsequent relaxation. The weights describe reference typicality, Π gives physical projection, and A contributes where agency is modelled. The local recording mechanism of Section 8 does not require J.

The point of AE is that these roles can belong to one architecture while remaining distinct. A successful model must state the bridge whenever it moves from one layer to another. That discipline is more important than the notation itself.

4 Rooted order, physical orientation and records

A rooted modal order matters physically only if its asymmetry can be transported through Π and leave records. This section states the required projection conditions, derives the corresponding orientation and record results, and marks the assumptions that remain physical postulates. The full argument and its comparison with rival accounts are developed in the companion paper on temporal directedness (Mury, 2026b).

4.1 What the root contributes

The root gives AE its most basic asymmetry: extensions are organised away from a common minimum. This does not yet mean that physical time has been derived. It means only that the modal structure distinguishes antecedence from extension.

The mathematics guarantees a least configuration in the abstract order. Figure 1 identifies that combinatorial minimum with a common physical anchor only under an additional projection/boundary assumption: Π must extend coherently to that anchor. AE adopts this identification in the present architecture, but it is a physical postulate, not a theorem of event-structure theory and not yet a derivation of a cosmological beginning.

4.2 Transporting an orientation

To test whether the rooted asymmetry has physical significance, take the directions in which histories extend away from an event and project them onto spacetime. The set K_x collects those projected directions at the physical point $x=\Pi(e)$.

$$K_x = \text{cone}\{d\Pi_e(v): \Pi(e) = x, v \in O_{\text{hist}}^+(e)\}.$$

Orientation-transport proposition. On a connected, time-orientable spacetime, suppose the projected extension directions are non-empty, non-zero and timelike at the relevant points and are jointly compatible with a continuous global time orientation. They then select that

orientation uniquely. Local agreement alone is not enough: global compatibility is an explicit hypothesis. The result transports a rooted asymmetry under a constrained projection; it does not create one from reversal-symmetric data (Mury, 2026b).

In plain language: AE can carry its root-to-extension direction into physical spacetime only if the projection respects the light-cone structure. The theorem therefore tells us what a successful physical implementation must preserve. It does not prove why nature has a root in the first place.

4.3 The record arrow

Records connect an oriented description with physically retained information. Let X label an earlier alternative and Y a later record. If each record value is compatible with at most m earlier alternatives, the conditional information loss is bounded by $\log m$. This is a property of the specified record map, not a derivation of thermodynamic entropy.

$$\text{Ent}(Y) \geq \text{Ent}(X) - \log m.$$

The equation is an ordinary information-theoretic bound. Since $\text{Ent}(X) \leq \text{Ent}(X,Y) = \text{Ent}(Y) + \text{Ent}(X|Y)$ and at most m values of X can remain compatible with a record value, $\text{Ent}(X|Y) \leq \log m$; hence $\text{Ent}(Y) \geq \text{Ent}(X) - \log m$. The argument also covers noisy records under the same multiplicity bound. If the record is one-to-one, it preserves all the information. This gives AE a precise record arrow without pretending that record information is already thermodynamic entropy (Reichenbach, 1956). The complete derivation and its role in temporal directedness are given in the temporal-directedness companion paper (Mury, 2026b).

5 Weights, action and structural entropy

Once orientation has been separated from thermodynamic dissipation, reference weights can be interpreted independently. Coherent weights behave consistently under refinement; multiplicative weights have an additive logarithmic representation; structural entropy concerns genuine alternatives. Physical entropy requires the additional state-space and access assumptions of Section 8.

5.1 Projective weights

If a coarse alternative is later divided into finer mutually exclusive alternatives, its reference weight must equal the sum of the reference weights of those finer alternatives. This is the consistency condition written in the next equation.

$$W(v) = \sum_{c \succ v} W(c), \quad W(c) = W(v)q(c | v).$$

In plain language, the children c of a genuine alternative v must exhaust the probability mass of v : the conditional weights $q(c|v)$ are non-negative and sum to 1. This additivity applies only when the children are mutually exclusive physical alternatives. Independent events that merely commute through a diamond must not be counted as separate exclusive outcomes.

Branching geometry alone does not fix reference weights. They require a physical or statistical justification in each model. In restricted dynamical classes, geometry or reduced flux may provide one; that possibility does not derive a universal law here. Until justified, w is additional fixed reference structure. Selection, including A where present, does not retrospectively alter it.

5.2 *The logarithmic sector of the action*

Suppose $W > 0$ is a multiplicative path weight and f is a continuous additive cost: $f(W_1 W_2) = f(W_1) + f(W_2)$, with $f(1) = 0$. The continuous logarithmic Cauchy equation then gives the form below. Requiring non-negative surprisal for $0 < W \leq 1$ fixes $k \geq 0$; choosing the usual probability normalisation fixes $k = 1$ (Aczél, 1966).

$$f(W) = -k \log W.$$

In plain language, the logarithm is what converts a product along a history into a sum of local contributions. If two successive alternatives have weights $1/2$ and $1/4$, their combined weight is $1/8$, while their surprisals add: $-\log(1/2) - \log(1/4) = -\log(1/8)$. This is why $-\log W$ appears naturally in one sector of J . The result does not prove that the entire functional J is negative log-probability or that J is universally a physical action (Aczél, 1966).

5.3 *Entropy of the alternatives*

At a finite cut, take a genuine partition of complete histories into mutually exclusive alternatives with weights $p(v)$, for v in Σ_n . Their structural entropy is the Shannon entropy written below. It measures how spread the reference weight is across the alternatives at that cut.

$$S_E(\Sigma_n) = - \sum_{v \in \Sigma_n} p(v) \log p(v).$$

If a genuine alternative is refined into finer exclusive alternatives, the entropy decomposes into the old entropy plus the average entropy created inside each parent alternative.

$$S_E(\Sigma_{n+1}) = S_E(\Sigma_n) + \sum_{v \in \Sigma_n} p(v) \text{Ent}(q(\cdot|v)).$$

Genuine branching can add distinctions, so the entropy of the alternative partition cannot decrease under refinement. But this is structural entropy over possibilities, not yet

Boltzmann entropy of matter. Counting duplicate descriptions would make the result meaningless, which is why the configuration quotient matters.

Agents enter at a different layer. The weights say how typical alternatives are under the fixed reference measure; agentic constraints may help determine which admissible history is selected, but they do not remodel those weights. AE therefore does not need an agent at the root in order to define structural entropy, and an agent's goal need not increase that entropy.

6 Agency within a complete history

AE places deliberation, goals, actions and their consequences inside the complete physical history. Completeness does not mean that an agent is causally ineffective: the history includes the effects of the agent's actions on other systems. A tenseless account represents those dependencies without a second time in which decisions are inserted into the block.

6.1 Functional agency and the stronger hypothesis

A functional agent requires more than a label attached to an event. Its model must specify a controller, a represented objective and actions that affect later physical records or states. A goal-sensitive evaluation can supply a finite consistency witness: with the event structure, projection and reference weights held fixed, changing the represented objective changes the compatible selected history. Such a witness establishes functional goal sensitivity in its stated model. It does not demonstrate that the agentic information is irreducible to physical information or to a matched stochastic account.

The stronger AE hypothesis is that an agentic variable *A* is not fixed by the relevant prior complete physical restriction, while having a physical expression somewhere in the complete admissible history. That is an additional hypothesis. If a complete physical state and closed deterministic dynamics already fix the physical continuation, merely adding the name *A* cannot create a different continuation. STIT approaches provide a useful comparison for representing richer notions of agency and choice (Belnap, Perloff, & Xu, 2001).

6.2 Selection and local causal efficacy

Where *A* enters a whole-history evaluator, it may change which admissible history is distinguished without changing the reference weights. The counterfactual reference measure remains the same after selection. Separately, within the realised history, an agent's physical actions can change another system's state through ordinary couplings. These are different roles: global compatibility does not replace the local interaction law, and local efficacy does not establish a global actuality rule.

Earlier finite selection constructions show that these roles can be represented consistently and that an agentic contribution can alter a winner under fixed non-agentic data. They remain supplementary witnesses, not the central mechanism of the recording

study. A plan bit or a logic gate used in a cascade calculation is a controlled component; it is not, by itself, a model meeting the stronger agency criterion.

6.3 What the theory does not assign to agency

A is not needed to create the root, construct spacetime, determine every reference weight or supply a primordial low-entropy preparation. Nor is A defined as an entropy-increasing force. An action can disperse, correlate or organise physical states; its entropy effect depends on the interaction, preparation, measure and macrodescription. The programme must derive that effect in a specified physical model rather than obtain it from the word agency.

7 Interactions and the composition of histories

AE must represent a world in which many systems act on one another. An agent may have a large downstream effect while another has little effect; a local change may be transmitted through several systems and feed back into the system that initiated it. These effects belong to a joint history with explicit physical couplings. The event structure organises their compatibility and dependencies; the physical realisation supplies what the interactions actually do.

7.1 Systems interacting within one history

Consider several systems with local records, controllers and shared environmental degrees of freedom. An action changes a physical state; a coupling transmits that change; another system records or responds to it; later actions may depend on those records. The joint history includes the whole sequence of dependencies. Heterogeneous coupling strengths and available actions explain why the same local change can have different effects at different sites. A cascade is a possible organisation of these effects, not an additional thermodynamic law.

This is the sense in which AE is explanatory and can have causal physical realisations. A concrete explanation specifies which extensions are admissible, which physical operations realise them and which later differences those operations produce. The existence of an event dependency alone does not specify an interaction strength, a Hamiltonian or a source distribution. Those physical ingredients must be given or derived independently.

7.2 Compatible developments composing

A local subsystem history is a restriction of a complete joint history. Two such restrictions can compose when they agree on shared events and states, preserve their prerequisites, avoid conflict and satisfy the relevant whole-history constraints. Local consistency alone does not establish global admissibility. Compatible developments may share a past, proceed independently and later interact. Once their outcomes enter a common coupling, their later developments need no longer be independent.

Independent events can also have several serial descriptions of the same configuration. Identifying those descriptions prevents an arbitrary ordering convention from producing extra alternatives, probability mass or entropy. Reconvergence is legitimate for compatible independent developments. It is not legitimate when it erases which incompatible past occurred. This distinction is central to the concurrent formulation of the global-admissibility study (Mury, 2026c).

7.3 *What composition does not mean*

In the minimal interpretation, mutually incompatible complete alternatives are not several concrete worlds exchanging signals. Comparing them in J is not a physical interaction between them. Constraints shared by several alternative histories are constraints on admissibility, not a flow of energy from an unrealised branch into the actual one. A theory of direct influence between incompatible alternatives would require an additional coupling law and an account of their physical status; the present mechanism does not supply it.

The phrase histories compound is therefore precise when it refers to compatible local histories composing into a joint history, or to effects propagating among systems inside that history. It must not silently change into an assertion of cross-world interaction. The distinction keeps the intuitive picture of intertwined developments compatible with AE's single actual complete history.

7.4 *Cascades as tests of composition*

Finite coupled-component witnesses test unequal influences, combinations of plans, downstream effects and feedback. They demonstrate that the AE bookkeeping can describe such composition and permit controlled comparisons with uncoupled or replayed dynamics. Their entropy results remain conditional on the stated initial sector and macrovolume. In the tested environmental witness, coupling did not universally increase growth and the imposed low environmental sector was not derived from recording capacity. These controls prevent a cascade from becoming an unsupported explanation of primordial low entropy. The five-component construction and controls are archived at [https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/computational/five_agent_cascades_v1/\(model.py and results.json\)](https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/computational/five_agent_cascades_v1/(model.py%20and%20results.json)), with definitions in [proofs/paper-c-modal-rarity/integrated_mechanism_results.md](#), Section 3, at the same commit.

8 **Physical entropy and local recording**

The entropy question concerns physical macrovolumes and histories, not merely the number of drawn branches. Three distinctions are necessary: a structural orientation is different from an entropy gradient; a reference distribution over alternatives is different from physical phase-space volume; and typical growth in an ensemble is different from what a particular actuality rule selects.

8.1 From physical states to macrovolumes

Boltzmann entropy uses the physical measure of the macroregion containing a state. A macrodescription therefore requires a physical evaluation, a reference state space and a partition. The spacetime projection Π does not itself supply all three. Modal rarity constrains physical rarity only under a separately controlled measure-transfer map; even then, occupying a small region does not prove that the realised trajectory later relaxes.

The current recording module makes the partition concrete through a reader. A collective coupling that records memory occupation groups configurations with the same occupation into one cell. A reader with access to all individual memory values instead distinguishes configurations that the collective reader groups together. Retaining the full writing trace changes the object again: it distinguishes histories by their recorded past. Instantaneous memory macrovolume and the information retained about a complete history are not interchangeable quantities.

8.2 Readiness constrained by a task

An exact recording task asks one fixed device, with one fixed preparation, to preserve a source and copy every allowed input into its memory. In the reversible XOR module, this universal capacity forces the memory to be blank before writing. The restriction is obtained from what the apparatus must accomplish, rather than from an entropy reward in J . More general finite capacity bounds account explicitly for auxiliaries.

This is a necessity conditional on an apparatus and a task. It does not show how the apparatus was built, why the task is instantiated in nature, or how the ready state was physically produced. It also does not transfer automatically from memory to every environmental register. Explaining the origin and cost of readiness is a further physical question.

8.3 Growth in the same prepared module

With the same blank memory, a declared source of independent balanced bits and reversible sequential writing, the occupation partition yields a typical increase of memory macrovolume over the studied late window. Exceptional histories are retained and bounded; the claim is not monotonic increase at every writing step. The mechanism does not require J . The result therefore connects functionally constrained preparation with typical macroscopic growth in one local apparatus, rather than joining two unrelated witnesses.

The recording study also constructs an effective reversible collective reader and states the permitted controls and record resources. These physical restrictions explain why occupation is the observable in that module. Allowing individual access or preserving a finer trace changes the partition and can remove the reported gain. The result is an entropy of the declared memory macrodescription, not an increase of fine-grained joint entropy, a dissipated heat calculation or a cosmological boundary theorem (Mury, 2026d).

8.4 *Actuality remains a separate obligation*

A statistical law whose conditional density is bounded relative to the prepared reference law can preserve a typical growth statement, with an explicit allowance for failed preparation. An unrestricted argmin need not do so: it may choose an exceptional or frozen history. Conversely, a rule can force low preparation without giving the subsequent trajectory an entropy gradient. Selection, preparation and growth thus require compatible hypotheses, not one interchangeable argument.

J concerns comparison of complete alternatives and, with a specified selection rule, the distinguished history. It is not the final event of a physical process and does not create the structural orientation. The root-to-extension order and its projection can orient histories whether or not an entropy observable increases along the chosen orientation. No external observer is needed to state the distinction.

9 **The logic of the research programme**

The foundational paper specifies the architecture and its interpretation. The sector papers test distinct explanatory obligations using explicit assumptions. Their results are preparatory in a precise sense: they provide constraints and constructive mechanisms that a common physical realisation must satisfy. They are already assessable results, but their conjunction is not yet a derivation of that common realisation.

9.1 *Physical orientation*

The orientation study separates time-orientability from selecting an orientation. Data exactly invariant under reversal cannot select one orientation through an intrinsic rule that respects that symmetry. Rooted extension supplies an asymmetry, and projected extension directions select an orientation when they are non-empty, timelike and globally compatible on the stated connected spacetime. The result gives a sufficient AE route and a restricted symmetry obstruction. It does not prove that physics uniquely requires AE's E and II, nor that the selected orientation is a direction of entropy increase (Mury, 2026b).

9.2 *Global admissibility*

The global-admissibility study asks what finite modal descriptions determine. Under separating and extendibility assumptions, they determine a coherent completion. They determine the admitted complete histories exactly when the admissible subset is closed: every excluded history must then have an exclusion witnessed by a finite restriction. Without that condition, all finite restrictions may remain extendible although a completed history is globally excluded. The criterion survives the identification of alternative serialisations of independent events.

The global-admissibility study therefore constrains what a reduced representation must retain. It does not derive reference weights, prove the need for a particular mixture of agents or determine actuality. A faithful formulation of AE in B-theoretic language must

retain the identified global information. The result is about information preserved by a representation, not a proof of a unique metaphysics (Mury, 2026c).

9.3 Recording and macroscopic entropy

The recording study asks whether a physically specified recording mechanism can link its task-constrained readiness to typical growth of an accessible macrovolume. It joins the capacity argument, sequential reversible writing, a declared source law and a collective physical readout. Event dependencies, projected operations and record resources identify when and how the relevant record can be obtained. A matched causal implementation preserving those ingredients realises the same local mechanism. The recording study supplies neither the global selection law nor the origin of the cosmological low-entropy boundary (Mury, 2026d).

9.4 How the papers constrain one another

The orientation study states what the projection must preserve to transport orientation. The global-admissibility study states what finite descriptions must preserve to represent global admissibility and concurrency. The recording study states what preparation, source, access and resource assumptions make one entropy mechanism work. A future common model must meet these conditions together with the same physical registers and independently fixed laws. Choosing an orientation model, a separate global constraint and an unrelated entropy device does not accomplish that unification.

The dependency is conceptual rather than a requirement that the recording study use J or that the global-admissibility study derive weights. The event structure organises admissible operations and alternatives; the projection gives their physical meaning; weights support typicality claims where needed; a selection rule answers a different question about actuality. The current local mechanism can be proved without deciding the full ontology of unrealised alternatives. It is a test of a shared physical module within AE, not a test of every AE commitment.

10 Explanatory value and comparison with existing theories

10.1 A causal realisation can express AE

AE is an explanatory theory whose physical modules may be causal. Describing an AE module causally is therefore not by itself a competing explanation. If an implementation preserves its admissible operations, dependency structure, projection, source law and resource constraints, it can be another representation of the same mechanism. This statement concerns that module; it does not turn every causal model into the whole AE theory.

AE's broader commitments include an objective organisation of admissible alternatives and an account of one complete actual history, with global evaluation in the variational formulation. These commitments are not required by the proof of every local module. Operational equivalence cannot decide their ontological interpretation. A comparator

may share the recording mechanism while interpreting its alternatives merely instrumentally, or while taking no position on a global actuality rule.

10.2 What would count as an advance

Branching structures, global constraints, reference measures, reversible recording and macrostate entropy have established precedents. AE's proposal is an explicit architecture of their roles, conditional links between those roles and constructive failure tests. Its stronger promise is that ingredients motivated independently for one task also restrict another: the same admissible extension structure supports orientation and access; the same physical projection constrains both; the same reference law supports several typicality questions.

For thermodynamics, the relevant background includes low-boundary and statistical-postulate accounts (Albert, 2000; Loewer, 2012). The recording study currently supplies a local task-relative route to readiness and growth, not a replacement for a cosmological Past Hypothesis. For tenseless laws, all-at-once and modal approaches already provide rich alternatives to purely generative laws (Adlam, 2022; Chen & Goldstein, 2022; Wharton, 2015). AE must be compared with those enriched accounts, not only with a bare realised manifold.

10.3 Tests against redescription

The comparison must state what is held fixed and what is removed. Erasing provenance may conflate incompatible pasts; counting serialisations may inflate weights; dropping a global condition may admit histories that finite data cannot exclude. In the recording module, changing access or retained records changes macrovolumes. These are assessable losses or changes. They show why the specified ingredients matter in their respective models, without establishing that each ingredient is exclusive to AE.

The stronger unification claim fails if E, Π , the source measure or the evaluator is retuned independently for each desired result. It gains support if a common physical specification fixes them before the targets are checked, reproduces the sector mechanisms and survives matched ablations. No new empirical signature or exhaustive priority claim follows from the current sector results. The framework offers a disciplined explanatory proposal; its comparative advantage remains a research question.

11 Outstanding physical bridges

The next programme stage is a common physically motivated model, not another repetition of the already established conditional recording calculation. It must state the physical origin of the relevant event dependencies and admissibility constraints, fix the projection without selecting the desired arrow in advance, and justify the source law, access restrictions and record resources in one setting. It should then test orientation and the local entropy mechanism without changing those ingredients between targets.

A separate production question concerns apparatus readiness: what physical process prepares a recorder, what other degrees of freedom carry its costs, and what boundary assumptions that process inherits. A separate actuality question concerns whether a proposed global rule preserves typical local growth rather than selecting its exceptions. Earlier rarity-transfer and gap arguments remain useful sufficient conditions in their stated model classes, but do not close either question by themselves. The rarity-transfer and stopping-line arguments are archived at https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/analytic_results.md, Sections 4–5; the separate positive-sector variational argument is in [proofs/paper-c-modal-rarity/variational_ph_proof.md](https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/variational_ph_proof.md) at the same commit.

The cosmological extension requires a gravitational and matter macrodescription, a justified measure, physical boundary conditions and a control of scale or regulator dependence. A small memory sector is not the low-entropy universe. Neither rootedness nor the capacity of a local recorder establishes cosmological initial conditions. Those bridges are future work, with premises and failure criteria to be stated independently.

The programme should preserve negative controls. Failure of a capacity constraint to restrict an environmental sector, a frozen selected history or disappearance of a gain under finer access identifies the scope of an explanation. Such findings are not repaired by relabelling entropy or agency. They determine which proposed bridge must be revised and which local results remain intact.

12 Conclusion

AE proposes one tenseless actual history together with an objective structure of physical alternatives. Dependence, conflict, independence and provenance organise compatible developments; a separately specified projection gives physical meaning to that structure. Systems interact within joint histories, while alternative complete histories remain alternatives. Reference weighting, agency and actuality retain distinct roles.

The current programme has conditional support for transport of orientation, a criterion for retaining global admissibility and a local recording mechanism linking task-constrained readiness with typical growth of a specified macrovolume. These results explain how particular AE components can work in physical realisations. They do not prove the whole ontology, produce cosmological initial conditions or establish an empirical advantage over every enriched causal or tenseless account.

The proposed contribution is a reusable architecture with explicit bridges and failure conditions. Its next decisive test is whether one independently motivated physical realisation can satisfy the sector constraints together. A causal formulation that preserves the same mechanism contributes to that test; a proof of distinct predictive or explanatory advantage over existing accounts requires further comparison.

Declaration on the use of generative AI

Generative AI systems, including OpenAI's ChatGPT and Codex and Anthropic's Claude, assisted conceptual exploration, adversarial review, formalisation, drafting, code development and structural revision. No AI system is credited as an author or treated as an evidential source. The ontological proposal and theoretical interpretation are those of the author.

Proof and source availability

The companion manuscripts contain the detailed sector arguments. The common public proof supplement is fixed to commit 0b30398bf80aa72fd47a040bf124080c9f4a1695: https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695. Precise archived paths are given in Sections 2.4, 7.4 and 11. Public availability of calculations does not imply public availability of these manuscripts. Private governance and exploratory logs are not premises of the arguments stated here.

References

- Aczél, J. (1966). *Lectures on Functional Equations and Their Applications*. Academic Press.
- Adlam, E. (2022). Laws of nature as constraints. *Foundations of Physics*, 52, Article 28. <https://doi.org/10.1007/s10701-022-00546-0>
- Albert, D. Z. (2000). *Time and Chance*. Harvard University Press. <https://doi.org/10.2307/j.ctvj57g>
- Belnap, N. (1992). Branching space-time. *Synthese*, 92(3), 385–434. <https://doi.org/10.1007/BF00414289>
- Belnap, N., Perloff, M., & Xu, M. (2001). *Facing the Future: Agents and Choices in Our Indeterminist World*. Oxford University Press. <https://doi.org/10.1093/oso/9780195138788.001.0001>
- Chen, E. K., & Goldstein, S. (2022). Governing without a fundamental direction of time: Minimal primitivism about laws of nature. In Y. Ben-Menahem (Ed.), *Rethinking the Concept of Law of Nature* (pp. 21–64). Springer. https://doi.org/10.1007/978-3-030-96775-8_2
- Everett, H. III. (1957). “Relative state” formulation of quantum mechanics. *Reviews of Modern Physics*, 29(3), 454–462. <https://doi.org/10.1103/RevModPhys.29.454>
- Le Bihan, B., Margoni, E., & Vieser, A. (2025). Possibility in physics. In H. Zwirn (Ed.), *Quantum Physics and Cosmology: The Mysteries of the Infinitely Small and the Infinitely Large* (Chapter 14, pp. 303–322). ISTE/Wiley. <https://doi.org/10.1002/9781394427673.ch14>
- Lewis, D. (1986). *On the Plurality of Worlds*. Basil Blackwell.
- Loewer, B. (2012). Two accounts of laws and time. *Philosophical Studies*, 160(1), 115–137. <https://doi.org/10.1007/s11098-012-9911-x>
- Mury, E. (2026b). Temporal Directedness Without an Open Future. Version 1. Preprint. 10.5281/zenodo.23206143
- Mury, E. (2026c). Global Constraints Beyond Finite Modal Data. Version 1. Preprint. 10.5281/zenodo.23256192
- Mury, E. (2026d). Recording Capacity and Macroscopic Entropy in Arborescent Eternalism. Version 1. Preprint. 10.5281/zenodo.23242803
- Nielsen, M., Plotkin, G., & Winskel, G. (1981). Petri nets, event structures and domains, part I. *Theoretical Computer Science*, 13(1), 85–108. [https://doi.org/10.1016/0304-3975\(81\)90112-2](https://doi.org/10.1016/0304-3975(81)90112-2)
- Reichenbach, H. (1956). *The Direction of Time* (M. Reichenbach, Ed.). University of California Press.
- Wallace, D. (2012). *The Emergent Multiverse: Quantum Theory According to the Everett Interpretation*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199546961.001.0001>
- Wharton, K. B. (2015). The universe is not a computer. In A. Aguirre, B. Foster, & Z. Merali (Eds.), *Questioning the Foundations of Physics* (pp. 177–189). Springer. https://doi.org/10.1007/978-3-319-13045-3_12